

Morphological Remodeling of Retinal Astrocytes in a Mouse Model of Ocular Hypertension

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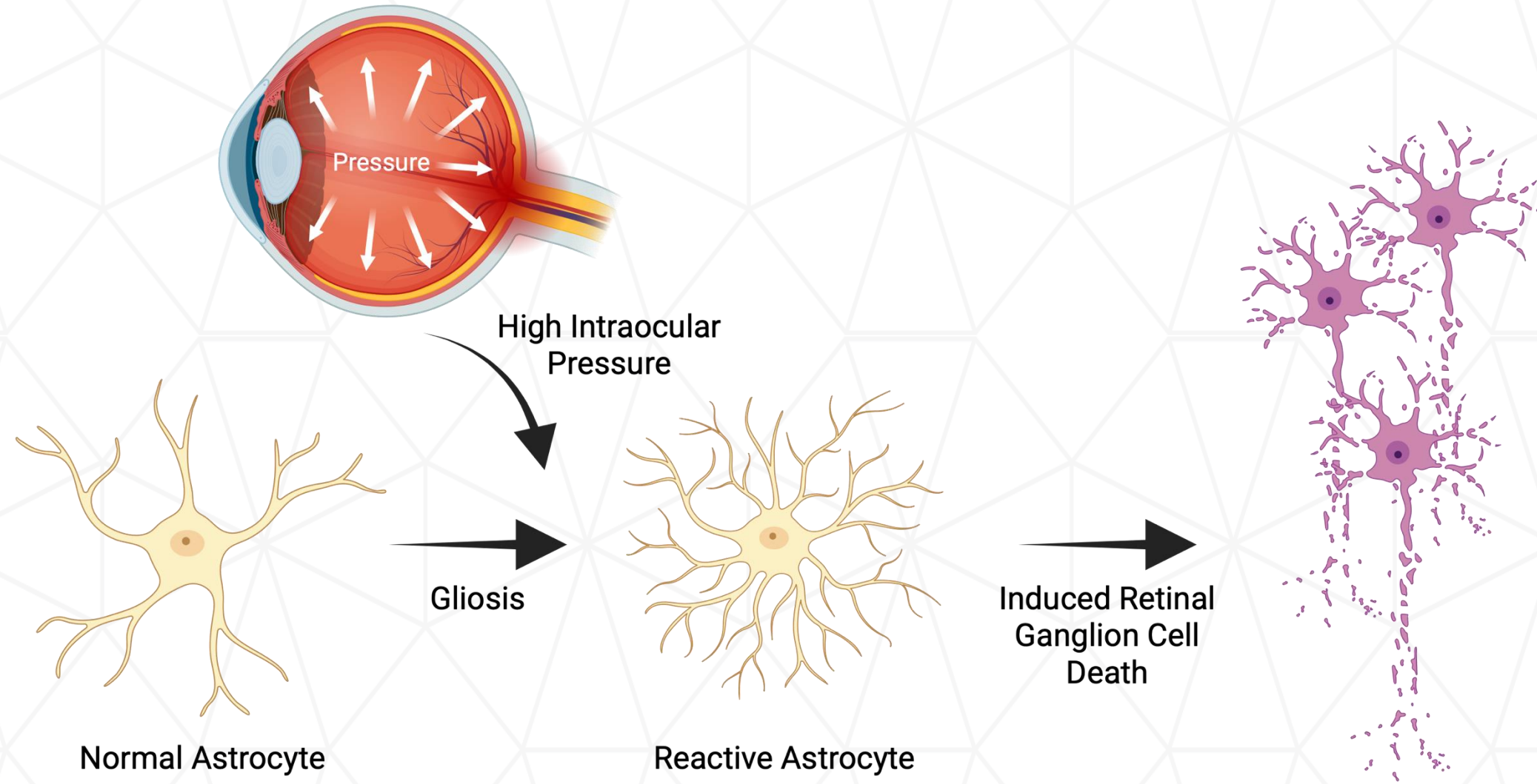
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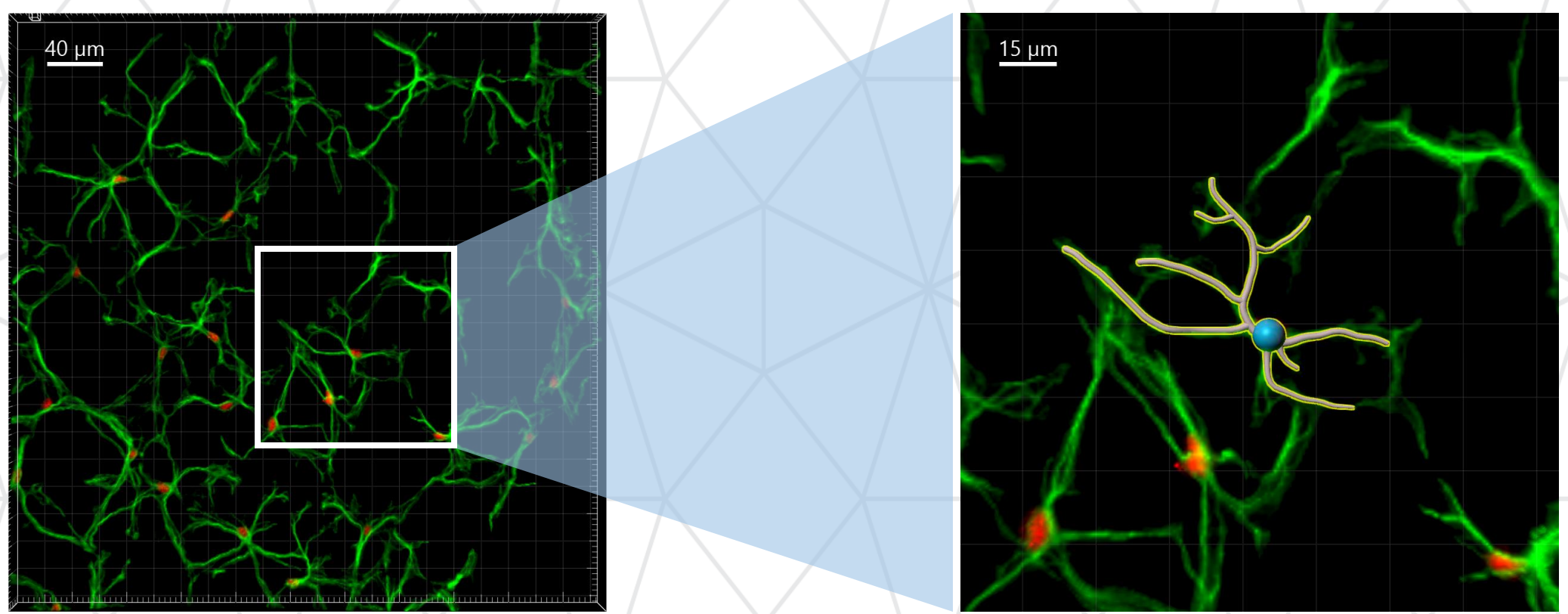
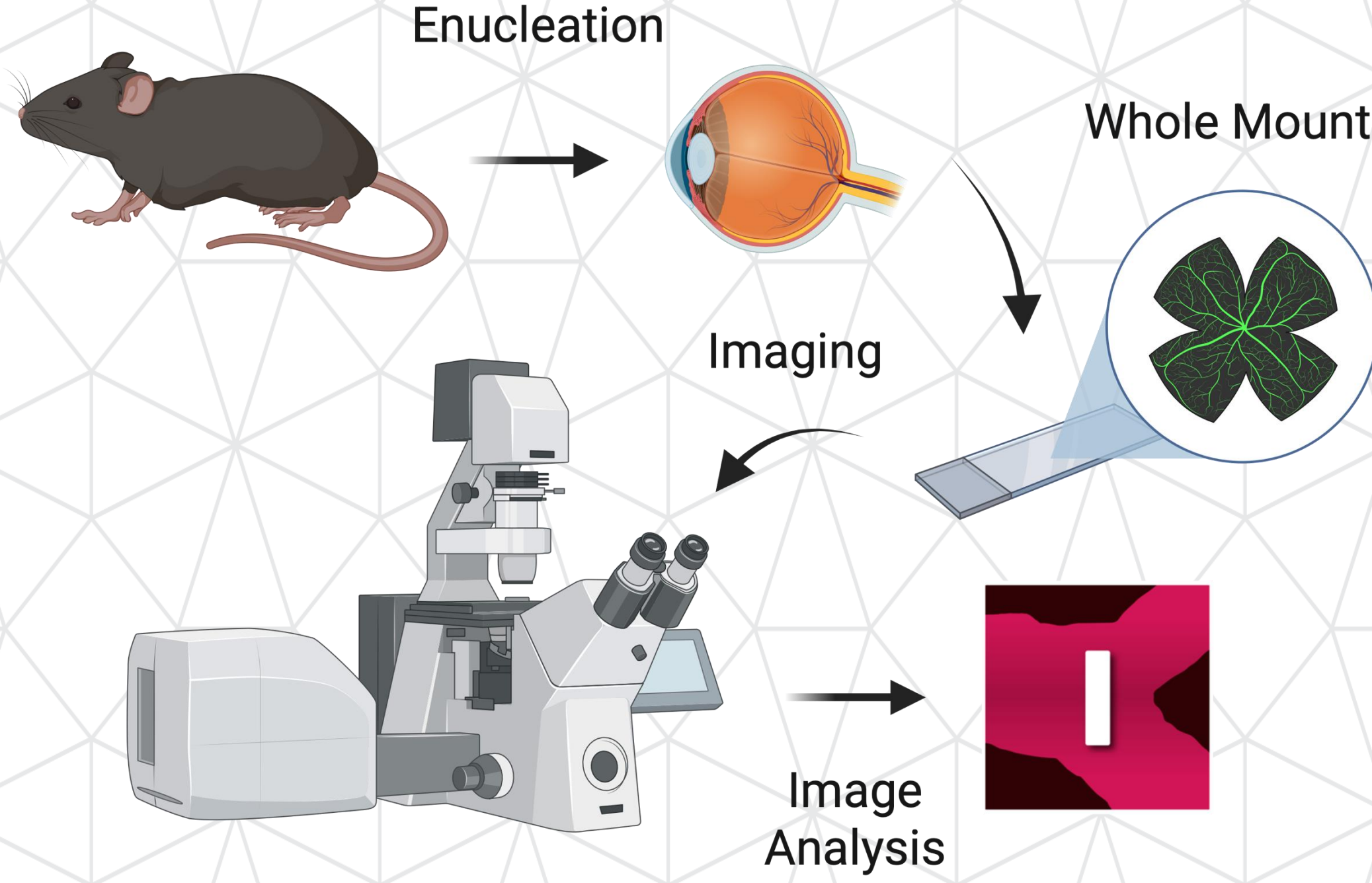
Introduction

- In a healthy retina, astrocytes maintain structural and metabolic support for ganglion cells¹.
- Under ocular hypertension, they exhibit morphological remodeling and molecular reprogramming, which may contribute to retinal ganglion cell degeneration in glaucoma^{1,2}.



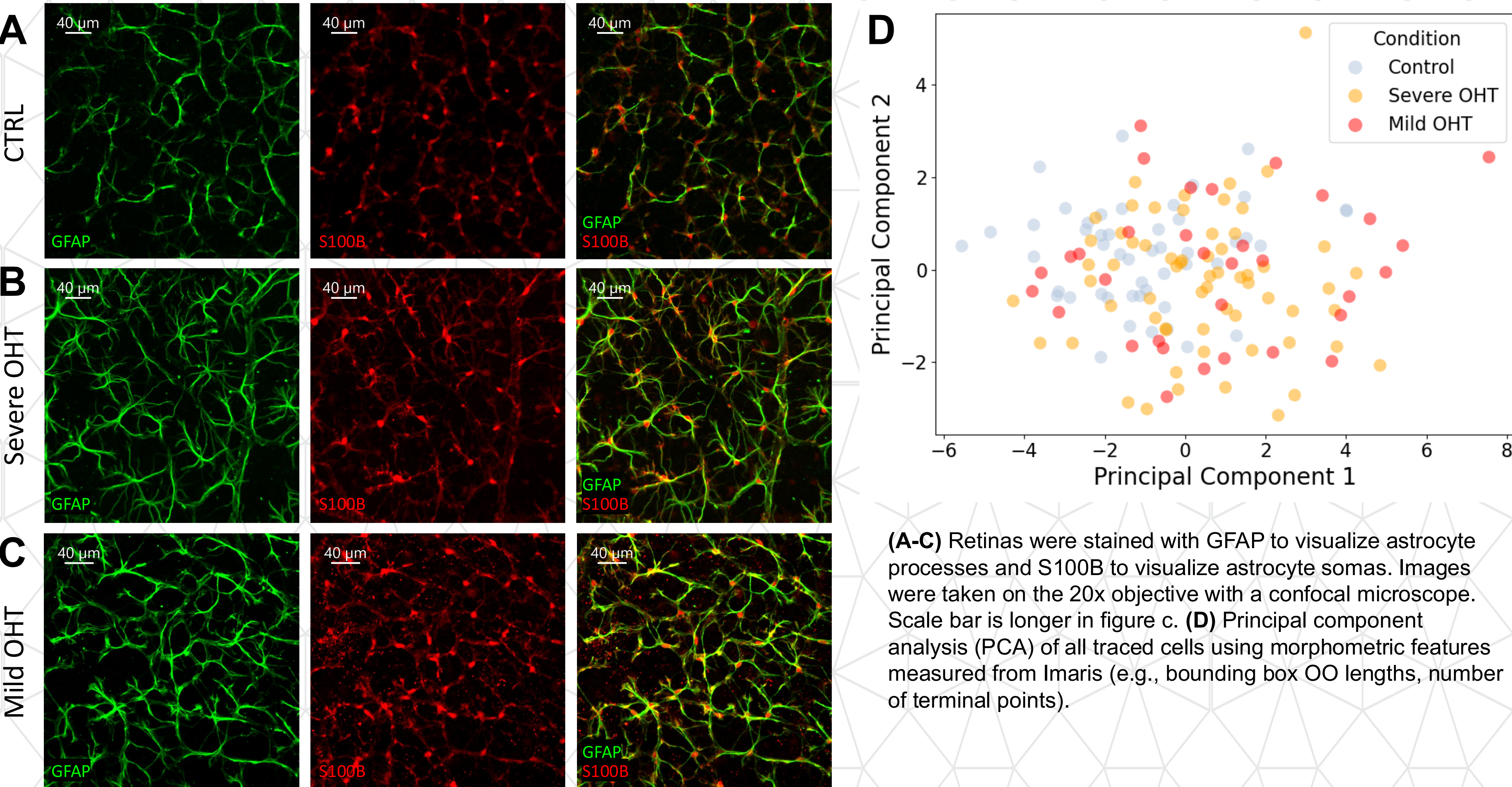
Aim: To characterize morphological remodeling of retinal astrocytes in both a severe and mild ocular hypertension (OHT) mouse model.

Methods



C57BL/6 mice were given a 1.8 μ L silicone oil injection in both eyes and harvested after 2 weeks for the severe OHT model. The mild OHT model was given a 1.2 μ L injection in both eyes and harvested after 8 weeks. Mice retina were immunostained with GFAP and S100B to visualize control and reactive astrocytes. Imaging was analyzed using Imaris v. 10.2.0 with its filament tracer.

Results



(A-C) Retinas were stained with GFAP to visualize astrocyte processes and S100B to visualize astrocyte somas. Images were taken on the 20x objective with a confocal microscope. Scale bar is longer in figure c. (D) Principal component analysis (PCA) of all traced cells using morphometric features measured from Imaris (e.g., bounding box OO lengths, number of terminal points).

Conclusions

Severe and mild OHT conditions exhibit vast morphological heterogeneity, even partially resembling the control morphology³.

- Severe OHT Model (2 weeks):** astrocytes show decreased length along a single axis and proximal process density without change to overall branching patterns or filament length⁴.
- Mild OHT Model (8 weeks):** astrocytes show increased total filament length and proximal process density while also exhibiting increased branching level and depth. Over eight weeks, astrocytes develop substantial morphological changes⁵.

Retinal astrocytes from severe and mild OHT models show differing morphologies. Future studies will examine transcriptomic characteristics of the two OHT models and investigate morphological and transcriptomic responses to a neuroprotective agent.

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